VIETNAM NATIONAL UNIVERSITY HANOI (VNU)
**VNU INFORMATION TECHNOLOGY INSTITUTE**

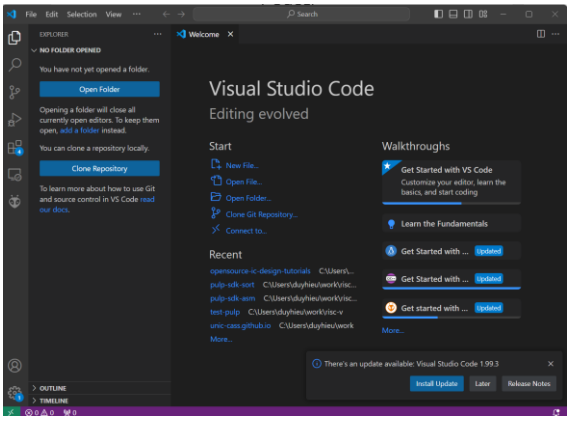
# Computer Architecture

# Install RISC-V simulator on VSCode

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https://duyhieubui.github.io

## Install VSCode

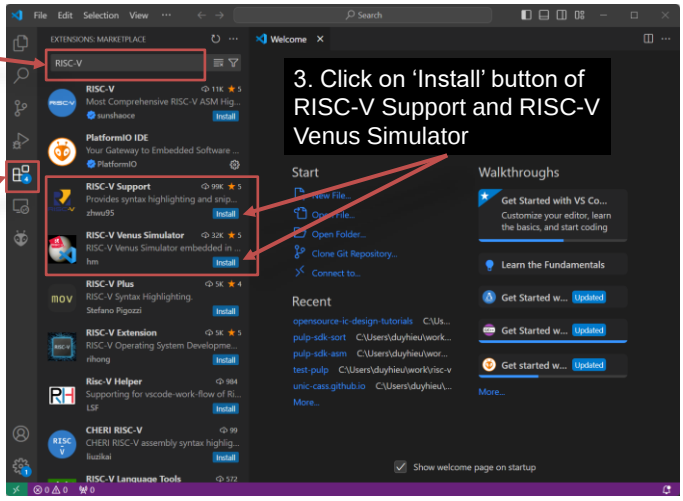
- Install Visual Studio Code by downloading and installing the corresponding files for your computer: https://code.visualstudio.com/



# Install RISC-V Venus Simulator

2. Search for RISC-V

1. Click on Market Place

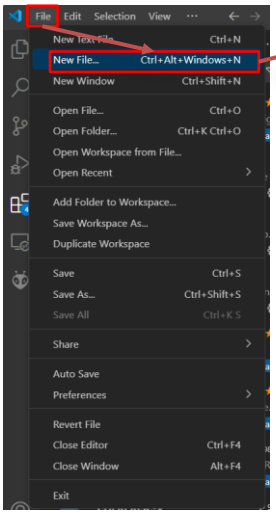


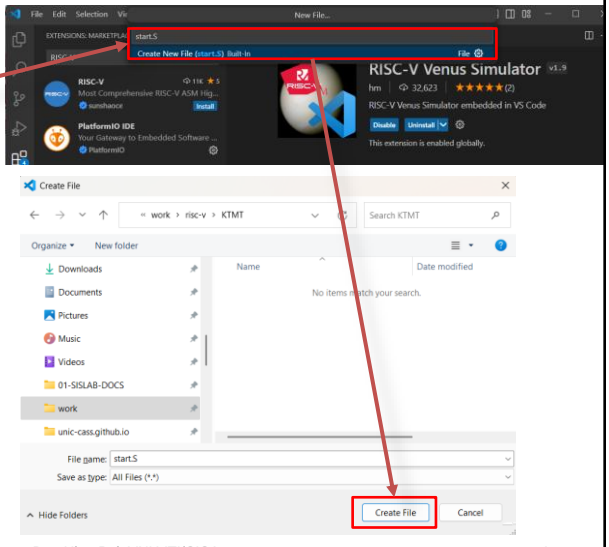

3. Click on 'Install' button of RISC-V Support and RISC-V Venus Simulator



# Create a new assembly file

- File >> New file

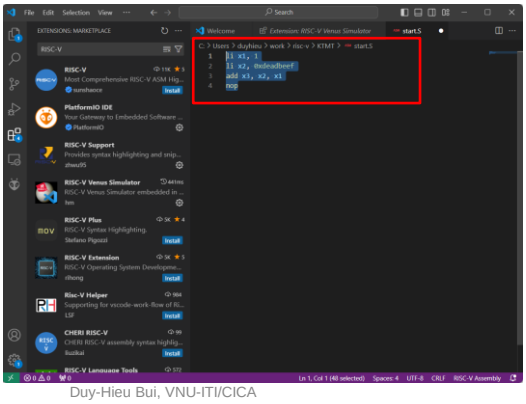


## Enter the RISC-V assembly instructions

- Enter the following instructions into the file

```
li x1, 1
li x2, 0xdeadbeef
add x3, x2, x1
nop
```

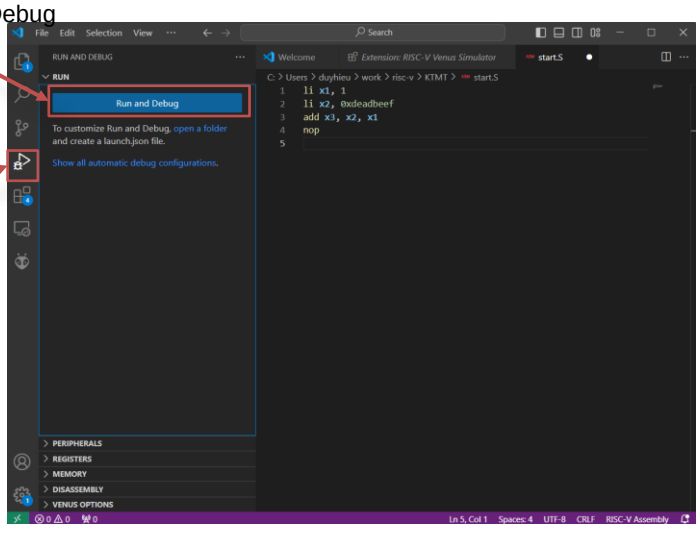




## Run the program

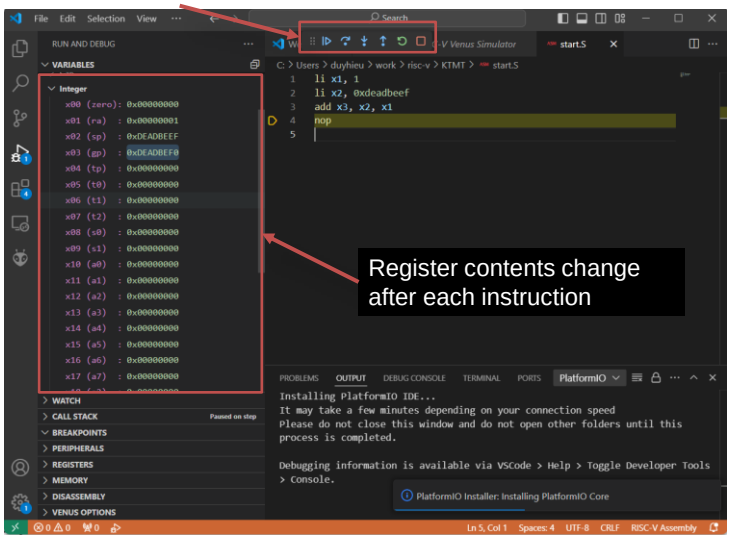
2. Click on Run and Debug

1. Click on Debug/Run icon



## Run instructions step by step

1. Click on the down arrow to run the next instruction






## Convert C/C++ into RISC-V assembly

- Open a web browser and go to https://godbolt.org
- Enter C/C++ into the source window
- Select “RISC-V (32-bits) gcc 14.12.0”

